

YEAR 7 • UNIT 2 • EXTRA HOMEWORK

One Letter at a Time

Expressions, substituting • and where volume comes from

Name: _____

Class: _____

Date given: _____

Date due: **Monday 14 September 2026****Why this sheet exists**

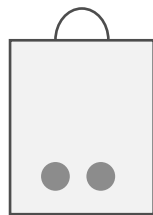
There is almost **no arithmetic** in Unit 2. Every number in it is small. What decides the answer is **one letter** — what it stands for, and where you put it.

So this sheet does one letter at a time, and it shows you the **middle line** of every calculation with the numbers left out. Filling in those boxes is not extra working. It **is** the method.

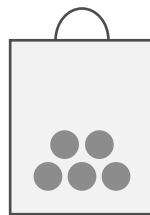
No calculator. Write in every box.

Part 1 • What a letter is**E1 The bag you cannot see into**

Write the number of counters in each bag. For the last one you cannot count them, so **choose your own letter** instead.



(a)



(b)



(c)

(a) counters(b) counters(c) counters**Why (c) needs a letter**

You cannot see inside bag (c), so nobody knows the number. That does not mean there is no number — there is one, and you just cannot count it. So we give it a **letter** instead, and carry on.

E2 Put three more in

Three counters are put into each bag. Fill in the boxes.

- (a) There are **6** counters in the bag.

Put in three more, so now there are $6 + 3 = \square$ counters.

- (b) There are **10** counters in the bag.

Put in three more, so now there are $\square + 3 = \square$ counters.

- (c) There are n counters in the bag.

Put in three more, so now there are $\square + 3$ counters.

- (d) There are n counters in the bag.

Take four out, so now there are $\square - \square$ counters.

Part (c) is the one that matters

You **cannot** work out $n + 3$. There is nothing to work out — you do not know what n is. So you **leave it as it is**. $n + 3$ is the **finished answer**, and this is called an **expression**.

E3 Say it the short way

The three rules

$4 \times n$ is written **4n** — the number goes **in front**.

$a \times b$ is written **ab**.

$w \div 2$ is written **w over 2**, as a fraction.

The first one is done for you.

(a) $4 \times n = 4n$

(d) $3 \times a \times b = \underline{\hspace{2cm}}$

(b) $7 \times k = \underline{\hspace{2cm}}$

(e) $w \div 2 = \underline{\hspace{2cm}}$

(c) $m \times p = \underline{\hspace{2cm}}$

(f) $t \div 5 = \underline{\hspace{2cm}}$

E4 One beaker, four things to do to it

Mr Bowen has a beaker holding $b \text{ cm}^3$ of water. Nobody has measured it, so we call it b .

Which operation?

pours in more	add
pours some out	subtract
three beakers the same	multiply
half of it	divide by 2

- (a) He pours in 40 cm^3 more. How much is in the beaker now? _____
- (b) He pours out 25 cm^3 . How much is left? _____
- (c) He has **three** beakers, all the same. How much altogether? _____
- (d) He pours away **half** of the first beaker. How much is left? _____

E5 Match the description to the expression

Every description uses the **same** two numbers, 3 and 8. Only the **order** changes. Write the correct **roman numeral** in each box. One expression is left over. The first one is done for you: **a** goes with **i**.

	Description			Expression
a	Multiply x by 3, then add 8.	i	i	$3x + 8$
b	Multiply x by 8, then add 3.		ii	$8x - 3$
c	Multiply x by 3, then subtract 8.		iii	$3 - 8x$
d	Multiply x by 3, then subtract the result from 8.		iv	$8x + 3$
e	Multiply x by 8, then subtract 3.		v	$3x - 8$
f	Multiply x by 8, then subtract the result from 3.		vi	$8 - 3x$
			vii	$8x - 8$

The one word to watch

subtract 8 means take 8 away from what you have.
subtract from 8 means **start** at 8 and take away what you have.
 Those two give different answers. The word is **from**.

Part 2 · Putting a number back in

E6 Expression or formula?

One difference, and it is the equals sign

An **expression** has letters and **no** equals sign: $7k$, $m + 9$.

A **formula** has letters **and** an equals sign: $A = lw$, $y = 3x$.

Write **E** for expression or **F** for formula. The first two are done for you.

- | | | | |
|--------------|----------|-------------------|-------|
| (a) $7k$ | E | (e) $5p - 2$ | _____ |
| (b) $y = 3x$ | F | (f) $V = 4t$ | _____ |
| (c) $m + 9$ | _____ | (g) $2a + 3b$ | _____ |
| (d) $A = lw$ | _____ | (h) $P = 2l + 2w$ | _____ |

E7 The same expression, four times

Only the number changes. Work down each list. The first one in each is done for you.

(a) The value of $m + 7$

when $m = 1$: $1 + 7 = 8$

when $m = 2$: $\square + 7 = \square$

when $m = 5$: $\square + 7 = \square$

when $m = 9$: $\square + 7 = \square$

(b) The value of $4m$

when $m = 1$: $4 \times 1 = 4$

when $m = 2$: $4 \times \square = \square$

when $m = 5$: $4 \times \square = \square$

when $m = 9$: $4 \times \square = \square$

Look at column (b) again

$4m$ means $4 \times m$. The times sign was hidden, not deleted.

So $4m$ with $m = 2$ is $4 \times 2 = 8$. **It is never 42.**

E8 The middle line, with the numbers left out

Every middle line below is already written for you. All you have to do is fill in the boxes, left to right.

(a) $5n$ when $n = 7$: $5n = 5 \times \square = \square$

(b) $n + 12$ when $n = 7$: $n + 12 = \square + 12 = \square$

(c) $3x + 4$ when $x = 6$:

$3 \times \square + 4 = \square + 4 = \square$

(d) $20 - 2n$ when $n = 8$:

$20 - 2 \times \square = 20 - \square = \square$

(e) $\frac{k}{4}$ when $k = 36$:

$\square \div 4 = \square$

(f) $2a + 3b$ when $a = 5$, $b = 2$:

$2 \times \square + 3 \times \square = \square + \square = \square$

Parts (c) and (d)

In both of them the **times** happens before the + or the -. That is why the middle line has three steps and not two – the multiplying is a step of its own, and it comes first.

E9 True or false – and if it is false, put it right

The first one is done for you.

Statement	True or false?	If false, the correct value is
(a) The value of $3m$ when $m = 4$ is 34.	False	$3 \times 4 = 12$
(b) The value of $2p$ when $p = 9$ is 18.		
(c) The value of $n + 5$ when $n = 11$ is 16.		
(d) The value of $6k$ when $k = 3$ is 63.		
(e) The value of $\frac{w}{3}$ when $w = 21$ is 7.		
(f) The value of $4 + 2t$ when $t = 5$ is 30.		

If a statement is false, **write the middle line** in the last column, the way you did in **E8**. That is where the mistake shows.

E10 Two formulae, one rectangle**The two rules for a rectangle**

Area: $A = lw$ – that is $l \times w$.

Perimeter: $P = 2l + 2w$ – the distance all the way round the edge.

Fill in the table. The first row is done for you.

l	w	$A = lw$	$P = 2l + 2w$
5	3	$5 \times 3 = 15$	$10 + 6 = 16$
8	2		
10	6		
7	7		

Part 3 · Where volume comes from

In Science you measured water in a cylinder, in cm^3 . Here is where that little **3** comes from.

Count the lengths you multiplied

One length: cm

A length **times** a length: $\text{cm} \times \text{cm} = \text{cm}^2$

A length **times** a length **times** a length: $\text{cm} \times \text{cm} \times \text{cm} = \text{cm}^3$

The small number is just **how many lengths you multiplied**. Nothing else.

And the unit that joins the two subjects: $1 \text{ cm}^3 = 1 \text{ ml}$, $1000 \text{ cm}^3 = 1 \text{ litre}$.

E11 Count the letters, then write the unit

Every letter stands for a length in centimetres. The first **two** rows are done for you.

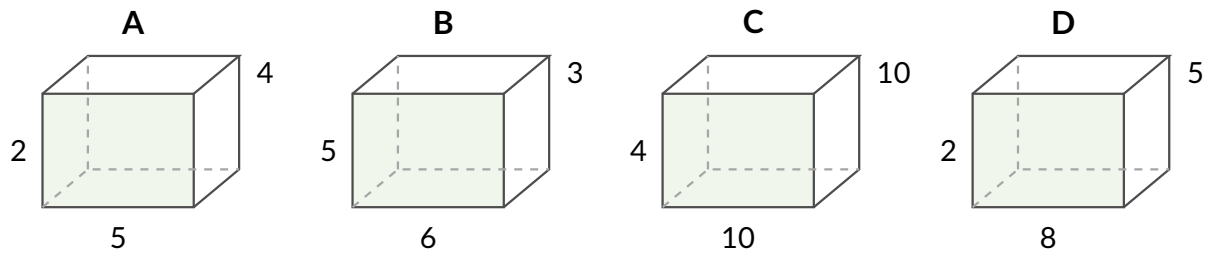
Expression	How many lengths multiplied?	Unit
h	one	cm
lw	two	cm^2
lwh		
$3ab$		
$s \times s \times s$		

E12 Find the volume of each box

One formula

$V = lwh$ means length $\times$ width $\times$ height. Three measurements, multiplied.

The first one is done for you.



$$\mathbf{A} \quad V = 5 \times 4 \times 2 = 40 \text{ cm}^3$$

$$\mathbf{B} \quad V = \square \times \square \times \square = \square \text{ cm}^3$$

$$\mathbf{C} \quad V = \square \times \square \times \square = \square \text{ cm}^3$$

$$\mathbf{D} \quad V = \square \times \square \times \square = \square \text{ cm}^3$$

E13 The tank, one step at a time

A tank has a rectangular base measuring **20 cm** by **10 cm**. Mr Bowen pours in water to a depth of h centimetres. Nobody has measured the depth, so we call it h .

(a) First, the **base**. Its area is $20 \times 10 = \square \text{ cm}^2$

(b) Now the water. Its volume is the base area **times** the depth:

$$V = \square \times h$$

(c) Write that the short way, with the number in front of the letter:

$$V = \underline{\hspace{2cm}}$$

(d) Now use it. When $h = 3$:

$$V = \square \times 3 = \square \text{ cm}^3$$

(e) And when $h = 7$:

$$V = \square \times \square = \square \text{ cm}^3$$

E14 Cubic centimetres, millilitres and litres

Science measures water in cm^3 and a bottle is labelled in ml. They are the same thing.

Two facts, and that is all

$1 \text{ cm}^3 = 1 \text{ ml}$ — exactly the same amount, two names.

$1000 \text{ cm}^3 = 1 \text{ litre}$

(a) $250 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ ml}$

(c) $1500 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ litres}$

(b) $2000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ litres}$

(d) $3 \text{ litres} = \underline{\hspace{2cm}} \text{ cm}^3$

(e) A water bottle holds 500 ml. A measuring cylinder in the science lab is marked in cm^3 . How many cm^3 of water does the bottle hold? _____

(f) The tank in **E13** holds $200h \text{ cm}^3$ of water. Mr Bowen fills it to a depth of $h = 25 \text{ cm}$. How many **litres** is that?

E15 Say what changed

Mr Bowen pours the **same** 200 cm^3 of water from a tall narrow cylinder into a wide flat dish. Nothing is added and nothing is spilled.

Tick one box in each row.

	Changes	Stays the same
How long the water is		
How wide the water is		
How deep the water is		
The volume of the water		

Now finish the sentence. Use each word **once**: **shape** **volume**

A liquid keeps the same _____ but takes the _____ of its container.